

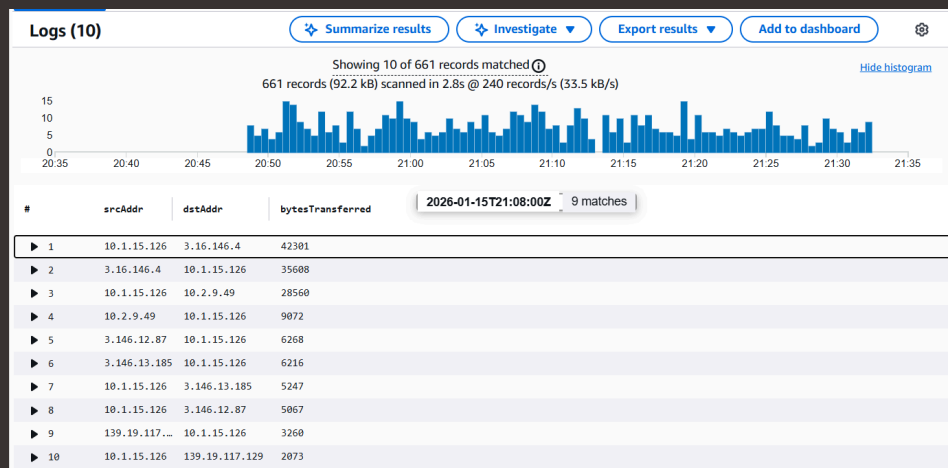


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VPC Monitoring with Flow Logs



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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a service that lets you create your own private network inside AWS where you can run cloud resources like EC2 instances. It is useful because it gives you full control over networking and security.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to connect and communicate between two instances in two different VPCs.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how cloud and data analysis work hand in hand to turn raw infrastructure logs into actionable insights about network traffic, security, and system performance. In other words, the cloud collects the data at scale, and data analysis lets you visualize patterns, detect anomalies, and make decisions quickly.



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This project took me...

This project took me 50 minutes to complete.



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In the first part of my project...

Step 1 - Set up VPCs

In this step, we will create two VPC's.

Step 2 - Launch EC2 instances

In this step, I will launching instances to allow the sharing of data information between them.

Step 3 - Set up Logs

In this step, I will set up a way to track all inbound and outbound network traffic and also set up a sapce that stores all of these records.

Step 4 - Set IAM permissions for Logs

I Will be giving VPC Flow Logs the permission to write logs and send them to cloudWatch and finish setting up the flow log.



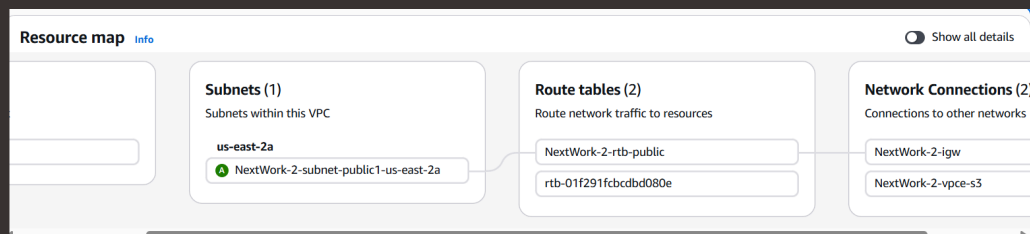
Multi-VPC Architecture

I started my project by launching two VPCs with two public subnets all together.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16 respectively. They have to be unique to avoid overlap of addresses of the resources in both VPCs.

I also launched EC2 instances in each subnet

My EC2 instances' security groups allow all ICMP traffic from ALL IP addresses. This is because we are trying to set up a connection between two VPCs.

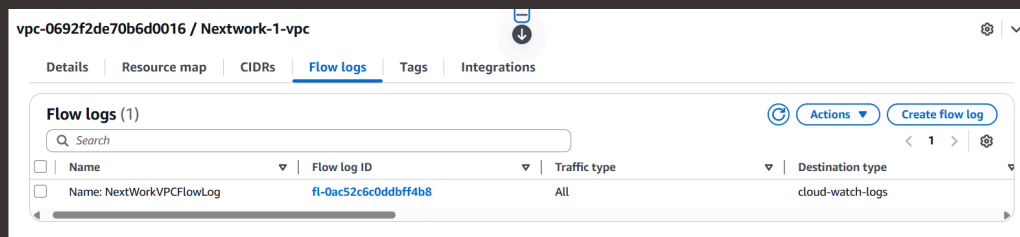




Logs

Logs are records that stores all events and activities such as logging in and errors encountered. Logs are important for monitoring, troubleshooting, security, and auditing

Log groups are like containers that manages and stores logs that are related .





IAM Policy and Roles

I created an IAM policy because by default, VPC Flow Logs don't have the permission to record logs and store them in your CloudWatch log group.

I also created an IAM role because policies need roles to be able to be attached to resources and services.

A custom trust policy is a policy that is attached to an IAM role which defines the services allowed to assume the role.

Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Sid": "Statement1",  
6       "Effect": "Allow",  
7       "Principal": {  
8         "Service": "vpc-flow-logs.amazonaws.com"  
9       },  
10      "Action": "sts:AssumeRole"  
11    }  
12  ]  
13 }  
14
```



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In the second part of my project...

Step 5 - Ping testing and troubleshooting

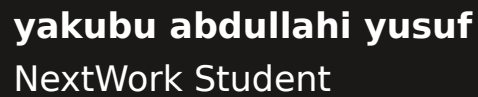
In this instance, I will try try get EC2 1 to send test messages to EC2 instance 2.

Step 6 - Set up a peering connection

In this step, I will set up a connection link between my VPCs by Creating a Peering Connection and Configuring Route Tables.

Step 7 - Analyze flow logs

In this step, I will be reviewing the flow logs recorded about VPC 1's public subnet and analysing the flow logs to get some insights.



```
#  
###  
  
#####  
#####  
#####  
#####  
#####  
#####  
#####  
#####  
#####  
#####
```

Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

```
[ec2-user@ip-10-1-15-126 ~]$ ping 10.2.9.49  
PING 10.2.9.49 (10.2.9.49) 56(84) bytes of data.
```

I could receive ping replies if I ran the ping test using the other instance's public IP address, which means Instance 2 is correctly configured to respond to ping requests, and Instance 1 can actually communicate with Instance 2 if it traffic goes across the public internet

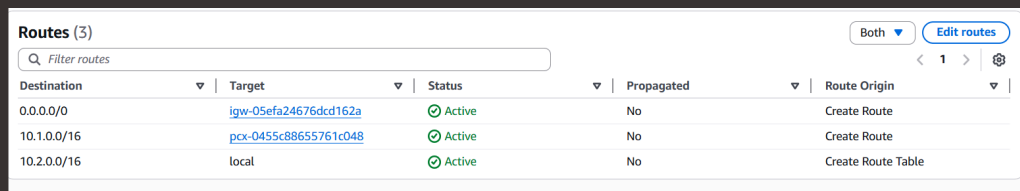


Connectivity troubleshooting

Looking at VPC 1's route table, I identified that the ping test with Instance 2's private address failed because VPC 1's route table had no route to VPC 2's CIDR through a VPC peering connection, so traffic to Instance 2's private IP had nowhere to go.

To solve this, I set up a peering connection between my VPCs

I also updated both VPCs' route tables so that traffic between the two VPCs can now flow directly through the peering connection instead of going through the internet.



The screenshot shows the AWS Route Tables console for a specific VPC. It displays three routes in a table. The first route is for destination 0.0.0.0/0, targeting the Internet Gateway (igw-05efa24676dcd162a). The second route is for destination 10.1.0.0/16, targeting the VPC Peering Connection (pcx-0455c88655761c048). The third route is for destination 10.2.0.0/16, with a target of 'local'. All three routes are in an 'Active' state. The 'Propagated' column shows 'No' for all routes. The 'Route Origin' column shows 'Create Route' for the first two and 'Create Route Table' for the third. There are search and filter options at the top, and pagination controls showing page 1 of 1.

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-05efa24676dcd162a	Active	No	Create Route
10.1.0.0/16	pcx-0455c88655761c048	Active	No	Create Route
10.2.0.0/16	local	Active	No	Create Route Table



Connectivity troubleshooting

I received ping replies from Instance 2's private IP address! This means that the VPC peering connection and route tables are correctly configured, and private IP communication between the instances is working.

```
[ec2-user@ip-10-1-15-126 ~]$ ping 10.2.9.49
PING 10.2.9.49 (10.2.9.49) 56(84) bytes of data:
64 bytes from 10.2.9.49: icmp_seq=1 ttl=127 time=0.516 ms
64 bytes from 10.2.9.49: icmp_seq=2 ttl=127 time=0.209 ms
64 bytes from 10.2.9.49: icmp_seq=3 ttl=127 time=0.212 ms
64 bytes from 10.2.9.49: icmp_seq=4 ttl=127 time=0.202 ms
64 bytes from 10.2.9.49: icmp_seq=5 ttl=127 time=0.210 ms
64 bytes from 10.2.9.49: icmp_seq=6 ttl=127 time=0.193 ms
64 bytes from 10.2.9.49: icmp_seq=7 ttl=127 time=0.224 ms
64 bytes from 10.2.9.49: icmp_seq=8 ttl=127 time=0.205 ms
64 bytes from 10.2.9.49: icmp_seq=9 ttl=127 time=0.209 ms
64 bytes from 10.2.9.49: icmp_seq=10 ttl=127 time=0.194 ms
64 bytes from 10.2.9.49: icmp_seq=11 ttl=127 time=0.210 ms
64 bytes from 10.2.9.49: icmp_seq=12 ttl=127 time=0.231 ms
64 bytes from 10.2.9.49: icmp_seq=13 ttl=127 time=0.196 ms
64 bytes from 10.2.9.49: icmp_seq=14 ttl=127 time=0.214 ms
64 bytes from 10.2.9.49: icmp_seq=15 ttl=127 time=0.214 ms
64 bytes from 10.2.9.49: icmp_seq=16 ttl=127 time=0.231 ms
64 bytes from 10.2.9.49: icmp_seq=17 ttl=127 time=0.217 ms
64 bytes from 10.2.9.49: icmp_seq=18 ttl=127 time=0.209 ms
```



Analyzing flow logs

Flow logs tell us about the network traffic going to and from network interfaces in a VPC.

For example, the flow log I've captured tells us that the instance with ENI eni-03e38dfb7843c5de4 sent 2 TCP packets from 10.1.15.126:42228 to 3.146.13.185:443, totaling 128 bytes, and the traffic was accepted.

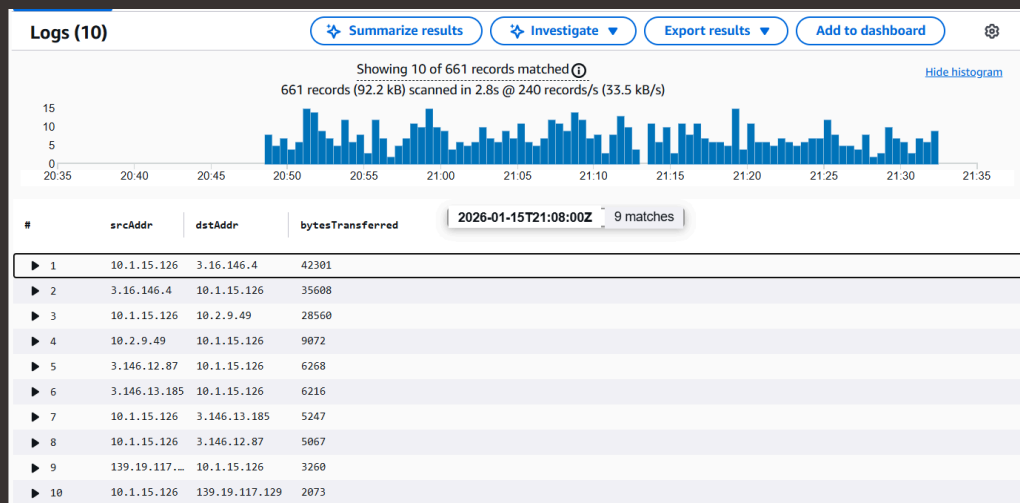
```
2026-01-15T20:51:42.000Z      2 176971016201 eni-03e38dfb7843c5de4 10.1.15.126 3.146.13.185 42228 443 6 2 128 1768510302 1768510327 ACCEPT OK
2 176971016201 eni-03e38dfb7843c5de4 10.1.15.126 3.146.13.185 42228 443 6 2 128 1768510302 1768510327 ACCEPT OK
2026-01-15T20:51:42.000Z      2 176971016201 eni-03e38dfb7843c5de4 205.210.31.230 10.1.15.126 51657 28016 6 1 44 1768510302 1768510327 REJECT OK
2 176971016201 eni-03e38dfb7843c5de4 205.210.31.230 10.1.15.126 51657 28016 6 1 44 1768510302 1768510327 REJECT OK
2026-01-15T20:51:42.000Z      2 176971016201 eni-03e38dfb7843c5de4 45.142.154.87 10.1.15.126 58914 4080 6 1 44 1768510302 1768510327 REJECT OK
2026-01-15T20:51:42.000Z      2 176971016201 eni-03e38dfb7843c5de4 52.207.222.50 10.1.15.126 123 40263 17 1 76 1768510302 1768510327 ACCEPT OK
2026-01-15T20:51:42.000Z      2 176971016201 eni-03e38dfb7843c5de4 10.1.15.126 52.207.222.50 40263 123 17 1 76 1768510302 1768510327 ACCEPT OK
```




Logs Insights

Logs Insights is a CloudWatch feature that analyzes your logs.

I ran the query `stats sum(bytes) as bytesTransferred by srcAddr, dstAddr | sort bytesTransferred desc | limit 10`. This query analyzes the top 10 byte transfers by source and destination IP addresses.





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